

## PARAMETRIC MODELS FOR POINT PROCESSES

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### 1. Introduction

We want to provide parametric models for point processes which are not renewal processes. Three different families of models are proposed. The paper is an *outline* giving some relevant results and exploring possible approaches to some of the unsolved problems.

The easiest way to specify point processes is to specify  $X_i, i \in I$ , the process of distances between points. This can be done directly, in which case the problem is to provide parametric models for non-negative discrete parameter stochastic processes. It can also be done indirectly, as shown in section 3.1, on the basis of discrete parameter processes which admit *negative* values.

### 2. Three families of stochastic processes

#### 2.1 Autoregressive-moving average processes

The classical models for discrete parameter stochastic processes are the autoregressive and moving average processes and a mixture of the two, the autoregressive-moving average process. (see, e.g., Anderson. (1971)). Over the last few years these models have been forged into a powerful tool in applied statistics (Box and Jenkins, (1970)). The idea of using them as such, or suitably modified, in our problem is tempting.

The process  $Z_i, i \in I$ , is said to be a stationary autoregressive-moving average (ARMA) process of order (p,q) if it satisfies.

$$Z_i - \phi_1 Z_{i-1} - \dots - \phi_p Z_{i-p} = a_i - \theta_1 a_{i-1} - \dots - \theta_q a_{i-q} \quad ,$$

where the parameters  $\phi_1, \dots, \phi_p$  and  $\theta_1, \dots, \theta_q$  have to satisfy certain conditions (See, e.g., Box and Jenkins (1970) and Anderson (1971)) and the  $a_i$  are independently and identically distributed and have finite variance  $\sigma_a^2$ . For our purposes it is best to look at the moving average representation of  $Z_i$ ,

$$Z_i = \sum_{j=0}^{\infty} \psi_j a_{i-j} \quad ,$$

where  $\psi_0 = 1$  and the other  $\psi_j$  can be calculated from the  $\phi_j$  and  $\theta_j$  (See, e.g., Anderson (1971), section 5.8).

Gaussian ARMA (p,q) processes arise if the  $a_i$  are normally distributed.

We shall show in section 3.1 that this process as such can be used to specify the process of distances implicitly.

If one wants to use ARMA processes to specify the process of distances directly one must make  $Z_i$  non-negative, e.g. by postulating  $\psi_j \geq 0$  and a non-negative distribution for the  $a_i$ . For the cases which we considered, this leads to complicated distributions of  $Z_i$ .

Another approach is to consider non-negative processes which are closely related to the Gaussian ARMA model. This is done in sections 3.2 and 3.3 using processes described in sections 2.2 and 2.3.

## 2.2 The gamma process with ARMA covariance

In (1970) we introduced the gamma point process, for which the finite dimensional distributions are multivariate gamma with parameters  $\beta > 0$  and  $R(r)$ ,  $r \in I$ . This process has mean  $\beta R(0)$  and covariance function  $\beta^2 R^2(r)$ . The model is only of practical value if  $R(r)$ , which has to be symmetric and positive definite, depends on a small number of parameters. The special case which was investigated in (1970) was  $R(r) = ae^{-\lambda|r|}$ ,  $0 < a, \lambda < \infty$ , the gamma point process with exponential covariance. To achieve more flexibility in this family of models one could postulate suitable further families of functions  $R(r)$ , depending on a few parameters.

Let  $Z_{ij}, i \in I, j = 1, 2, \dots, p$  be  $p$  independent stationary Gaussian processes with zero expectations, then  $X_i = \sum_{j=1}^p Z_{ij}^2$ ,  $i \in I$ , is Gamma with parameters  $\beta = p/2$  and  $R(r)$ , where  $R(r)$  is the covariance function of  $Z_{ij}$  (Linhart, Zucchini (1970)). One could now use as a family of models gamma processes  $X_i$ , which are derived in this way from Gaussian ARMA processes. One arrives at gamma processes with a flexible family of parameter functions  $R(r)$ , which only depend on a few parameters. A generalisation would then immediately be necessary from  $\beta = p/2$ ,  $p$  integer, to arbitrary real positive  $\beta$ .

If  $R(r; \phi_1, \phi_2, \dots, \phi_p, \theta_1, \dots, \theta_q)$  is the covariance function of an ARMA (p,q) process and  $\beta > 0$ , then a gamma process with parameters  $\beta$  and  $R(r; \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)$  will be called a gamma process with ARMA (p,q) covariance.

The gamma process with exponential covariance, considered in Linhart (1970) and Linhart and Zucchini (1970), is the special case with ARMA (1,0) covariance.

## 2.3 The lognormal ARMA process

Another way to use Gaussian processes  $Z_i$  to arrive at non-negative processes  $X_i$  is to take

$$X_i = \exp(Z_i).$$

$X_i$  could then be called a lognormal process.

The finite dimensional p.d.f.'s of a stationary lognormal process  $X_i$  can be

obtained directly from the finite dimensional p.d.f.'s of a stationary Gaussian process  $Z_i$ .

$$f_{X_{i+1}, \dots, X_{i+n}}(x_1, x_2, \dots, x_n) = (1/(2\pi)^{n/2} |\Lambda|^{1/2} x_1 x_2 \dots x_n)$$

$$\exp \left( -(1/2) \sum_{j,k=1}^n \lambda^{jk} (\log x_j - \mu) (\log x_k - \mu) \right)$$

$$\text{for } 0 \leq x_j, j = 1, 2, \dots, n,$$

where  $\Lambda = (\lambda_{ij}) = (R(i-j))$ , and  $R(r)$  is the covariance function of  $Z_i$ .

By expanding  $\exp(i(u_1 X_{i+1} + \dots + u_n X_{i+n}))$  into a power series, and using the well known characteristic function of  $Z_{i+1}, Z_{i+2}, \dots, Z_{i+n}$ , one obtains the characteristic function of  $X_{i+1}, X_{i+2}, \dots, X_{i+n}$ .

$$\varphi_{X_{i+1}, \dots, X_{i+n}}(u_1, u_2, \dots, u_n) =$$

$$\sum_{i_1, i_2, \dots, i_n=0}^{\infty} (i_1^{i_1+i_2+\dots+i_n} u_1^{i_1} u_2^{i_2} \dots u_n^{i_n} i_1! i_2! \dots i_n!)$$

$$\exp. (\mu \sum_{j=1}^n i_j + (1/2) \sum_{j,k=1}^n R(j-k) i_j i_k).$$

Of interest is the mean of  $X_i$ , its covariance function, and its correlation function:

$$E(X_i) = \exp(\mu + R(0)/2)$$

$$R_X(r) = (\exp(R(r)) - 1) \exp(2\mu + R(0))$$

$$P_X(r) = (\exp(R(r)) - 1) / (\exp(R(0)) - 1),$$

(Mostafa and Mahmoud (1964)).

If  $Z_i$  is Gaussian ARMA(p,q), then  $X_i$  will be called lognormal ARMA(p,q).

### 3. The Structure of the associated point processes

It remains to be seen how successful the processes introduced in section 2 will be in applications. Decisive in this respect will be how complex the most important theoretical properties of the corresponding point processes are, and whether it is possible to devise efficient, and computationally manageable methods of estimating the

parameters of the process. We investigate the probabilistic structure of the three point processes in this section and make some remarks on the estimation problem in section 4.

### 3.1 The normal point process

A process of distances,  $X_i, i \in I, X_i \geq 0$ , can also be specified in the following way. Let  $Z_i, i \in I$ , be a stationary discrete parameter stochastic process with  $E(Z_i) > 0$ , where the  $Z_i$  can also take on negative values.

Let  $Y_0 = 0$

$$Y_i = Z_1 + Z_2 + \dots + Z_i \quad i = 1, 2, \dots$$

$$Y_i = -Z_0 - Z_{-1} - \dots - Z_{i+1} \quad i = -1, -2, \dots$$

and put

$$Y_{(0)} = Y_0 = 0$$

and order the  $Y_i$  according to magnitude, calling the ordered variables  $Y_{(i)}$ , that is

$$X_i = Y_{(i+1)} - Y_{(i)} \geq 0.$$

$X_i$  can then be used as process of distances. If the process  $Z_i$  is specified, so is  $X_i$ .

We conjecture that  $X_i, i \in I$ , is stationary, but have not been able to prove this. It seems difficult to obtain the finite dimensional distribution functions of  $X_i$ .

One can, however, get the intensity function of the point process.

If  $h(\tau)$  is the intensity function,  $h(\tau) d\tau$  is the probability to have a point in  $(\tau, \tau + d\tau)$ , given a point at zero. It can be seen that in our case

$$h(\tau) = \sum_{n=1}^{\infty} (f_{Y_n}(\tau) + f_{Y_{-n}}(\tau)),$$

where  $f_{Y_n}(\tau)$  is the p.d.f. of  $Y_n$ . This can also be written as

$$h(\tau) = \sum_{n=1}^{\infty} (f_{Y_n}(\tau) + f_{Y_n}(-\tau))$$

since, by stationarity of  $Z_i, i \in I$ ,  $Y_n$  and  $-Y_{-n}$  are identically distributed. (If  $Z_i$  is non-negative  $f_{Y_n}(-\tau) = 0$  for all  $\tau > 0$  and one obtains the familiar expression for the intensity function.)

In the case where  $Z_j$  is Gaussian with mean  $\mu$  and covariance function  $R(r)$ ,

$$f_{Y_n}(\tau) = (1/\lambda_n \sqrt{2\pi}) \exp(-(\tau - n\mu)^2 / 2\lambda_n^2), \quad n = 1, 2, \dots,$$

where

$$\lambda_n = \sum_{j=-(n-1)}^{(n-1)} (n - |j|) R(j).$$

Therefore

$$h(\tau) = \sum_{n=1}^{\infty} (1/\lambda_n \sqrt{2\pi}) (\exp(-(\tau - n\mu)^2 / 2\lambda_n^2) + \exp(-(\tau + n\mu)^2 / 2\lambda_n^2)).$$

It is easy to specialise this result for ARMA  $(p, q)$  processes by substituting the well known expressions for  $R(j)$ .

### 3.2 The gamma point process

The most important theoretical properties of the gamma point process are given in Linhart (1970). A drawback of the gamma point process is that the general form of the probability density functions of distances and waiting times are hopelessly complicated and that the intensity function is not explicitly known. Some further investigation whether the results simplify to a certain extent for gamma processes with ARMA covariance seems necessary. In particular one should try to give the characteristic function of the waiting times and the Laplace transform of the intensity function more explicitly.

We tackled this problem for the process with ARMA  $(1, 0)$  and  $(0, 1)$  covariance.

The Laplace transform of  $h(t)$ , the intensity function, of a gamma point process with parameters  $\beta$  and  $R(r)$  is

$$h^*(s) = \sum_{n=1}^{\infty} |I + s\Sigma_n|^{-\beta},$$

where  $\Sigma_n = (R(i - j))$ , (Linhart, 1970).

$$\text{Now } |I + s\Sigma_n| = \kappa^{-n} |\kappa I - \Sigma_n|,$$

where  $\kappa = -1/s$ . The coefficient  $t_i$  in the characteristic polynomial of  $\Sigma_n$

$$|\kappa I - \Sigma_n| = \kappa^n - t_1 \kappa^{n-1} + \dots + (-1)^n t_n$$

is the sum of all  $i$ -rowed principal minors of  $\Sigma_n$  (see, e.g., MacDuffee (1946), p. 19). There are  $\binom{n}{i}$   $i$ -rowed principal minors in  $\Sigma_n$  and all of them are equal to  $|\Sigma_i|$ . With this approach one gets explicit expressions for  $|I + s\Sigma_n|^{-\beta}$  the Laplace transform of the probability density function of the waiting times, and for  $h^*(s)$ .

The covariance function of an ARMA (1,0) process is

$$R(r) = \phi_1^{|r|} \sigma_a^2 / (1 - \phi_1^2).$$

For a gamma process with ARMA (1,0) covariance

$$\Sigma_n = (\sigma_a^2 \phi_1^{|i-j|} / (1 - \phi_1^2))$$

and

$$|\Sigma_n| = \sigma_a^{2n} / (1 - \phi_1^2)$$

The Laplace transform of the intensity function is then

$$h^*(s) = \sum_{n=1}^{\infty} ((1 + s\sigma_a^2)^n - \phi_1^2)^{-\beta}$$

For an ARMA (0,1) process, i.e. a moving average of order one,

$$R(0) = \sigma_a^2 (1 + \theta_1^2) \quad -1 < \theta_1 < 1$$

$$R(1) = -\theta_1 \sigma_a^2$$

$$R(r) = 0 \quad \text{for } r \geq 2$$

Here

$$|\Sigma_n| = \sigma_a^{2n} \sum_{k=0}^{[(n+2)/2]} (-1)^k \binom{n-k}{k} (1 + \theta_1^2)^{n-2k} \theta_1^{2k}$$

(Uppuluri and Carpenter (1969))

and

$$|K I - \Sigma_n| = \kappa^n - \sum_{i=1}^n (-1)^i \binom{n}{i} \kappa^{n-i} \sigma_a^{2i} \sum_{k=0}^{[(i+2)/2]} (-1)^k \binom{i-k}{k} (1 + \theta_1^2)^{i-2k} \theta_1^{2k}$$

The Laplace transform of  $h(t)$  is then

$$h^*(s) = \sum_{n=1}^{\infty} (1 + \sum_{i=1}^n \binom{n}{i} (s\sigma_a^2)^i \sum_{k=0}^{[(i+2)/2]} (-1)^k \binom{i-k}{k} (1 + \theta_1^2)^{i-2k} \theta_1^{2k})^{-\beta}$$

Similar results could be obtained for processes with higher order ARMA covariance, once the determinants  $|\Sigma_n|$  have been calculated.

### 3.3 The lognormal point process

From the characteristic function of the joint distribution of  $X_{i+1}, X_{i+2}, \dots, X_{i+n}$ , given in section 3, follows immediately the characteristic function of the waiting times to the  $n$ th event,  $W_n = X_{i+1} + \dots + X_{i+n}$

$$\varphi_{W_n}(u) = \sum_{i_1, i_2, \dots, i_n=0}^{\infty} ((iu \exp(\mu))^{i_1+i_2+\dots+i_n} / i_1! \dots i_n!) \exp\left(\frac{R(0)}{2} \sum_{j=1}^n i_j^2 + \sum_{j=1}^{n-1} R(j) \sum_{k=1}^{n-j} i_k i_{k+j}\right).$$

In the case of ARMA (1,0) covariance the Laplace transform of  $h(\tau)$  is then

$$h^*(s) = \sum_{n=1}^{\infty} \sum_{i_1, i_2, \dots, i_n=0}^{\infty} ((-se^{\mu})^{i_1+i_2+\dots+i_n} / i_1! i_2! \dots i_n!) \exp\left(\frac{\sigma_a^2}{1-\phi_1^2} \left(\frac{1}{2} \sum_{j=1}^n i_j^2 + \sum_{j=1}^{n-1} \phi_1^j \sum_{k=1}^{n-j} i_k i_{k+j}\right)\right)$$

In the ARMA (0,1) case it becomes

$$h^*(s) = \sum_{n=1}^{\infty} \sum_{i_1, i_2, \dots, i_n=0}^{\infty} ((-se^{\mu})^{i_1+i_2+\dots+i_n} / i_1! i_2! \dots i_n!) \exp\left(\sigma_a^2 \left(\frac{1+\theta^2}{2} \sum_{j=1}^n i_j^2 - \theta_1 \sum_{k=1}^{n-1} i_k i_{k+1}\right)\right).$$

Thus, although the probability density function of the joint distribution of distances of a lognormal point process can be explicitly given, some of the other results are by no means simpler than those for the corresponding gamma point process.

## 4. Estimation

### 4.1 Normal point processes

The problem here is to estimate the parameters of  $Z_i, i \in I$ , from observations on  $X_i, i \in I$ .

If the coefficient of variation of  $Z_i$  is small, in most cases the  $Y_{(i)}$  will be the same as the  $Y_i$  and therefore the  $Z_i$  the same as the observed  $X_i$ . The problem is therefore reduced to estimating the parameters of  $Z_i$  from observations of  $Z_i$ . In the case where the  $Z_i$  are Gaussian ARMA (p,q), many suitable methods are available (Box and Jenkins (1970)).

If inversions are frequent, estimation methods must be based on the process  $X_i$ . If the joint distribution of the  $X_i$  is manageable one could use the method of maximum likelihood. The problem could also be approached via the estimation of the intensity

function, which depends on the parameters of the  $Z_1$ .

#### 4.2 *Gamma point processes with ARMA covariance*

One could either use, as approximations, exact maximum likelihood methods for Gaussian processes. Or, again, use Whittle's (1952) asymptotic maximum likelihood method, as is Linhart (1970). This method is an asymptotic maximum likelihood method for Gaussian processes but can be regarded as a method of least squares for arbitrary stationary processes.

Exact maximum likelihood methods for gamma processes are impossible because of the difficult joint density of such processes but, and this is the third possibility, one could use methods of estimation based on the *Laplace transform* of the joint density, which is relatively simple for the gamma process. Leslie (1970) presented recently a paper on such a method for the case where the observations  $X_1, X_2, \dots, X_n$  are realisations of  $n$  independently and identically distributed random variables. The methods would have to be generalised to the case where the  $n$  variables are not independently distributed.

#### 4.3 *Lognormal ARMA processes*

Here one has, to transform the observations to logarithms and then to apply techniques for Gaussian processes, which in this case are very thoroughly worked out (Box and Jenkins (1970)).

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